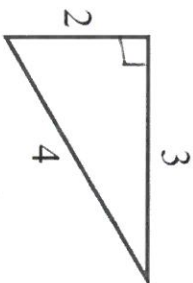


Lesson 1

You Do

We know that if the triangle has a right angle, then $c^2 = a^2 + b^2$. The converse of this is that if $c^2 = a^2 + b^2$ then the triangle must have a right angle. Test if $c^2 = a^2 + b^2$ to see if these triangles must have a right angle. They may not be drawn to scale.

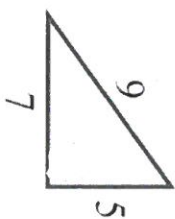
a



$$3^2 + 4^2 = 5^2$$

$$9 + 16 = 25 \quad \checkmark$$

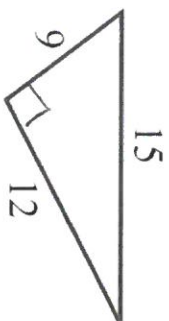
b



$$7^2 + 5^2 = 9^2$$

$$49 + 25 = 81 \quad \text{No}$$

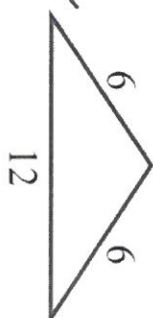
c



$$9^2 + 12^2 = 15^2$$

$$81 + 144 = 225 \quad \checkmark$$

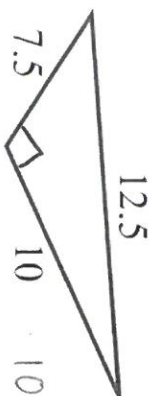
d



$$6^2 + 6^2 = 12^2$$

$$36 + 36 = 81 \quad \text{No}$$

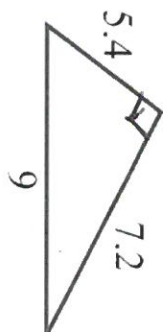
e



$$7.5^2 + 10^2 = 12.5^2$$

$$56.25 + 100 = 156.25 \quad \checkmark$$

f



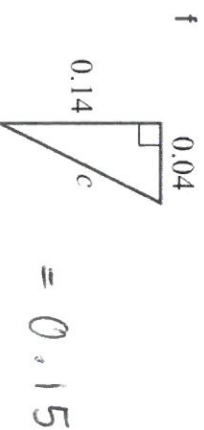
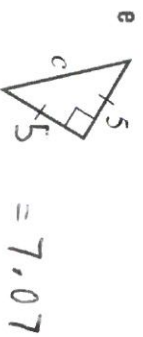
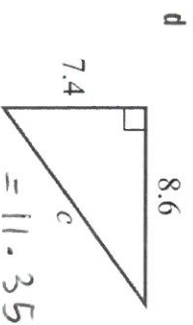
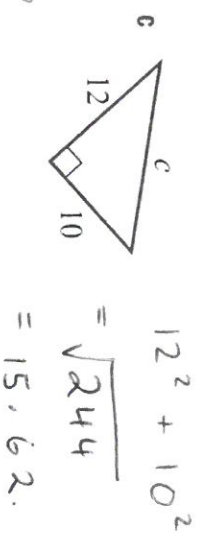
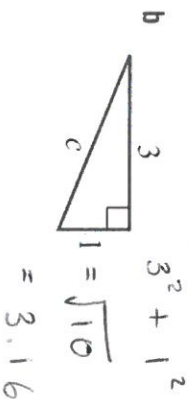
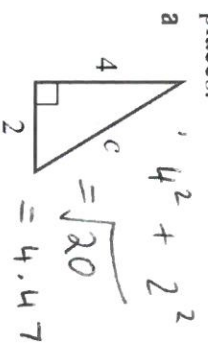
$$5.4^2 + 7.2^2 = 9^2$$

$$29.16 + 51.84 = 81 \quad \checkmark$$

$\therefore$ a, c, e, f are all right angled triangles

Practice

- 5 Find the length of the hypotenuse in each of these right-angled triangles, correct to two decimal places.



- 6 Find the length of the hypotenuse in these triangles, leaving your answer as an exact value.

